Wei Wang

Ph.D. Candidate in Mathematics, School of Mathematical Sciences, Peking University

wwmath166@outlook.com Website Google Scholar ORCID

Education

Sept. 2022 – 2027 (expected) **Peking University, School of Mathematical Sciences**
Ph.D. Candidate in Mathematics
Beijing, China
Advisor: Prof. Zhifei Zhang

Sept. 2018 – June 2022 **Zhejiang University, School of Mathematical Sciences**
B.S. in Mathematics and Applied Mathematics
Hangzhou, Zhejiang, China
Advisor: Prof. Ting Zhang
Thesis: *Homogenization Problem of Second Order Elliptic Systems with Lower Order Terms*

Research Interests

Geometric variational problems; elliptic and parabolic partial differential equations; harmonic maps and their flows; liquid crystal theory; homogenization theory; nonlocal operators.

Publications and Accepted Papers

1. Sharp convergence rates for parabolic Green's functions in time-independent periodic homogenization.
Wei Wang. Accepted for publication in *Frontiers of Mathematics*. [arXiv: 2606.12101]

2. Landau–de Gennes model with sextic potentials: asymptotic behavior of minimizers.
Wei Wang and Zhifei Zhang. Accepted for publication in *Memoirs of the European Mathematical Society*. [arXiv: 2404.00677]

3. Existence of weak solutions for two-phase matrix-valued harmonic map flows.
Wei Wang, Wei Wang (王伟), and Zhifei Zhang. *Journal of Functional Analysis* 291 (2026), no. 11, Article No. 111623. [doi] [arXiv: 2506.16278]

4. Uniform estimates of Landau–de Gennes minimizers in the vanishing elasticity limit with line defects.
Haotong Fu, Huaijie Wang, and **Wei Wang**. *SIAM Journal on Mathematical Analysis* 58 (2026), no. 4, 3306–3324. [doi] [arXiv: 2508.01811]

5. Stratification and rectifiability of harmonic map flows via tangent measures.
Haotong Fu, **Wei Wang**, Ke Wu, and Zhifei Zhang. *Mathematische Annalen* 395 (2026), Article No. 92. [doi] [arXiv: 2504.14880]

6. On the sharp constants in the regional fractional Sobolev inequalities.
Rupert L. Frank, Tianling Jin, and **Wei Wang**. *SN Partial Differential Equations and Applications* 6 (2025), Article No. 15. [doi] [arXiv: 2403.00357]

7. Uniform estimates of resolvents in homogenization theory of elliptic systems.
Wei Wang. *Journal of Differential Equations* 370 (2023), 1–65. [doi] [arXiv: 2211.14614]

8. Homogenization theory of elliptic systems with lower order terms in dimension two.
Wei Wang and Ting Zhang. *Communications on Pure and Applied Analysis* 22 (2023), no. 3, 787–824. [doi] [arXiv: 2210.06329]

Preprints and Manuscripts

1. Uniform boundary observability for periodic waves, I: up to the ε^{-1} eigenvalue scale.
Jun Geng, Shumao Wang, and **Wei Wang**. Preprint. [arXiv: 2610.33547]

2. Asymptotics for two-dimensional parabolic vectorial Allen–Cahn systems.
Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**. Preprint. [arXiv: 2609.27596]

3. Small-energy regularity for stationary weak solutions of the Liouville equation.
Haotong Fu, Kelei Wang, **Wei Wang**, and Ke Wu. Preprint. [arXiv: 2609.26026]

4. Boundary asymptotics for two-dimensional vectorial Allen–Cahn systems.
Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**. Preprint. [arXiv: 2609.00739]

5. Boundary regularity for Oseen–Frank minimizers with arbitrary positive splay, twist, and bend constants.
Zhiyuan Dai, Haotong Fu, Huaijie Wang, and **Wei Wang**. Preprint. [arXiv: 2608.22287]

6. Sharp instability of planar isotropic–nematic interfaces in the Landau–de Gennes model.
Wei Wang and Qin Wu. Preprint. [arXiv: 2608.12131]

7. Smooth convergence of Landau–de Gennes minimizers in the Lyuksyutov regime.
Haotong Fu, Huaijie Wang, and **Wei Wang**. Preprint. [arXiv: 2608.08456]

8. On the absence of point defects in biaxial Landau–de Gennes models.
Haotong Fu, Huaijie Wang, **Wei Wang**, and Zhifei Zhang. Preprint. [arXiv: 2608.04908]

9. Singular limits for three-dimensional global minimizers of Ginzburg–Landau-type functionals: uniform estimates and singular sets.
Giacomo Canevari, Haotong Fu, and **Wei Wang**. Preprint. [arXiv: 2606.27691]

10. A note on an approximate fixed point theorem for VMO maps.
Haotong Fu and **Wei Wang**. Preprint. [arXiv: 2606.15670]

11. Asymptotics of minimizers for Ginzburg–Landau-type functionals in high dimensions.
Giacomo Canevari, Haotong Fu, and **Wei Wang**. Preprint. [arXiv: 2605.04442]

12. Improved convergence of Landau–de Gennes minimizers in the vanishing elasticity limit.
Haotong Fu, Huaijie Wang, and **Wei Wang**. Preprint. [arXiv: 2507.14955]

13. Fine structure of rupture set for semilinear elliptic equations with singular nonlinearity.
Wei Wang and Zhifei Zhang. Preprint. [arXiv: 2411.16048]

14. Quantitative stratification and sharp regularity estimates for supercritical semilinear elliptic equations.
Haotong Fu, **Wei Wang**, and Zhifei Zhang. Preprint. [arXiv: 2408.06726]

Research Visits and Programs

Mar. – Apr. 2026 — **University of Verona, Department of Computer Science**
Visiting Ph.D. Student
Verona, Italy
Host: Prof. Giacomo Canevari

June – Sept. 2021 — **HKUST, Department of Mathematics**
Summer Research Program (Online)
Hong Kong, China
Advisor: Prof. Tianling Jin

Selected Talks

Aug. 2026 — Singular Limits for Three-Dimensional Global Minimizers of Ginzburg–Landau-Type Functionals: Uniform Estimates and Singular Sets
Conference on Geometric Measure Theory and Partial Differential Equations
Enshi, Hubei, China.

July 2026 — Uniform Estimates for Landau–de Gennes Minimizers with Point and Line Defects
Academic Forum on Theory and Numerical Analysis of Liquid Crystals and Complex Systems
Xiangtan University.

Jan. 2026 — Landau–de Gennes Model with Sextic Potentials: Asymptotic Behavior of Minimizers
Academic Workshop on Fluid Dynamics Equations
Peking University.

Nov. 2025 Stratification for Harmonic Map Flows via Tangent Measures: Rectifiability and Regularity
Jinhua Conference on Harmonic Analysis and Its Applications
Zhejiang Normal University.

Aug. 2025 Improved Convergence of Landau–de Gennes Minimizers in the Vanishing Elasticity Limit
Conference on Theory and Computation of Quasi Crystals
Nankai University.

June 2025 Rectifiability of the Stratification for Harmonic Map Flows via Tangent Measures
PDE and Geometry Seminar
Zhejiang University.

Grants and Projects

2023 – 2028 **Physical Modelling and Mathematical Theory of Liquid Crystals**
Participant, NSFC, Grant No. 12288101.

2023 – 2028 **Physical Modeling and Mathematical Theory of Liquid Crystal Systems**
Participant, National Key R&D Program of China, Grant No. 2023YFA1008801.

Academic Service

- **Mathematical Reviews / MathSciNet, American Mathematical Society**
  Registered prospective reviewer.
- **Academic Committee, 4th Frontiers Doctoral Forum**
  Screening and organization of talks in Analysis and Differential Equations.

Teaching Experience

2022 – 2026 **Peking University, Teaching Assistant**
Advanced Mathematics (A, B, and C); Real Analysis; Functional Analysis (I); Harmonic Analysis.

2021 – 2022 **Zhejiang University, Teaching Assistant**
Mathematical Analysis (I and II).

Honors and Awards

Sept. 2026 **National Scholarship**, School of Mathematical Sciences, Peking University.

Apr. 2025 **First Prize, Graduate Research Exhibition**, School of Mathematical Sciences, Peking University.

Apr. 2024 **Third Prize, Graduate Research Exhibition**, School of Mathematical Sciences, Peking University.

Oct. 2023 **Elite Program for Graduate Students in Mathematics**, Peking University.

Aug. 2022 **Nomination Award, ICCM Creative Undergraduate Thesis**, Shing-Tung Yau Center of Southeast University.

Apr. 2021 **First Prize, 11th Chinese Mathematics Competitions Finals**, Wuhan University.

June 2021 **Nomination Award, S.-T. Yau College Student Mathematics Contest**, Yau Mathematical Sciences Center, Tsinghua University.

Skills and Languages

Skills LaTeX, MATLAB, Mathematica, Python, Lean.

Languages Chinese (native), English (fluent).